

1. A direct mapped cache is of size 32 KB and has block size 32 bytes. CPU also generates 32 bit address. Number of bits needed for indexing the cache:

Memory address = 32 bits

Data Cache = 32 KB = 2^{15} B

Block Size = 32 B = 2^5 B

$$\text{number of blocks in cache memory} = \frac{2^{15}}{2^5} = 2^{10}$$

So, the number of bits required to index the cache blocks = $\lceil \log_2 2^{10} \rceil = 10$

2. Consider a cache consisting of 128 blocks of 16 words each. Assume main memory is addressable by 16-bit address. Main memory has 64 K words, which will be viewed as 4K blocks of 16 words each. The following shows the format of memory address in different mappings.

Tag	Block	Word	
5	7	4	Main memory address

Figure 5.15 Direct-mapped cache.

Tag	Word	
12	4	Main memory address

Figure 5.16 Associative-mapped cache.

Tag	Set	Word	
6	6	4	Main memory address

Figure 5.17 Set-associative-mapped cache with two blocks per set.

3. In a 4-way set associative cache, the total cache size is 64 KB and block size is 16 bytes. What is the number of sets in the cache?

The number of sets in the cache is **1024**. GW School of Engineering & Applied Science

1. Calculate the total number of blocks

First, determine the total number of blocks (or lines) in the cache by dividing the total cache size by the block size. Ⓢ

- **Total Cache Size:** $64 \text{ KB} = 64 \times 1024 \text{ bytes} = 65,536 \text{ bytes}$.
- **Block Size:** 16 bytes.
- **Total Blocks:** $\frac{65,536 \text{ bytes}}{16 \text{ bytes/block}} = 4,096 \text{ blocks}$. GW School of Engineering & Applied Scien... +2

2. Determine the number of sets

In a set-associative cache, blocks are grouped into sets. The number of blocks per set is defined by the associativity (K). Ⓢ

- **Associativity (K):** 4 (since it is a 4-way set associative cache).
- **Number of Sets:** $\frac{\text{Total Blocks}}{\text{Associativity}} = \frac{4,096}{4} = 1,024 \text{ sets}$

GW School of Engineering & Applied Scien... +2

Result

The total number of sets in a 4-way set associative cache with a 64 KB capacity and 16-byte block size is 1024. Ⓢ

4. A system uses a direct-mapped cache with 512 blocks and a block size of 32 bytes. What is the size of the tag field for a 32-bit memory address?

The size of the tag field is 18 bits.

1. Calculate Block Offset bits

The block offset bits are determined by the block size. For a block size of 32 bytes, the number of bits needed to address each byte within that block is:

$$\text{Block Offset} = \log_2(\text{Block Size}) = \log_2(32) = 5 \text{ bits}$$

2. Calculate Index bits

In a direct-mapped cache, the index bits are determined by the number of blocks (also called lines) in the cache. For 512 blocks, the number of bits needed for the index is:

$$\text{Index Bits} = \log_2(\text{Number of Blocks}) = \log_2(512) = 9 \text{ bits}$$

3. Calculate Tag bits

The total memory address is divided into three fields: Tag, Index, and Block Offset. For a 32-bit address, the tag field size is the remaining bits after accounting for the index and offset:

$$\text{Tag bits} = \text{Total Address bits} - (\text{Index bits} + \text{Block Offset bits})$$

$$\text{Tag bits} = 32 - (9 + 5) = 32 - 14 = 18 \text{ bits}$$

Result

The size of the tag field for a 32-bit memory address is 18 bits.

5.

6. In a 4-way set associative cache, the total cache size is 64 KB and block size is 16 bytes. What is the number of sets in the cache?

The number of sets in the cache is **1024**.

1. Calculate the total number of blocks

First, determine the total number of blocks (or lines) in the cache by dividing the total cache size by the block size.

- **Total Cache Size:** $64 \text{ KB} = 64 \times 1024 \text{ bytes} = 65,536 \text{ bytes}$.
- **Block Size:** 16 bytes.
- **Total Blocks:** $\frac{65,536 \text{ bytes}}{16 \text{ bytes/block}} = 4,096 \text{ blocks}$.

2. Determine the number of sets

In a set-associative cache, blocks are grouped into sets. The number of blocks per set is defined by the associativity (K).

- **Associativity (K):** 4 (since it is a 4-way set associative cache).
- **Number of Sets:** $\frac{\text{Total Blocks}}{\text{Associativity}} = \frac{4,096}{4} = 1,024 \text{ sets}$

Result

The total number of sets in a 4-way set associative cache with a 64 KB capacity and 16-byte block size is 1024.